

STAR Robot Control System

Comprehensive Technical Documentation

Complete Reference for Software Engineers

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1 Nanopb (Protobuf) Protocol with ARQ and CRC for SPI Communication

1.1 Executive Summary

Overview: Protocol Buffers (nanopb) encapsulated in frames with CRC-32 and Automatic Repeat Request (ARQ) for reliable SPI communication between RPi5 and ESP32-S3.

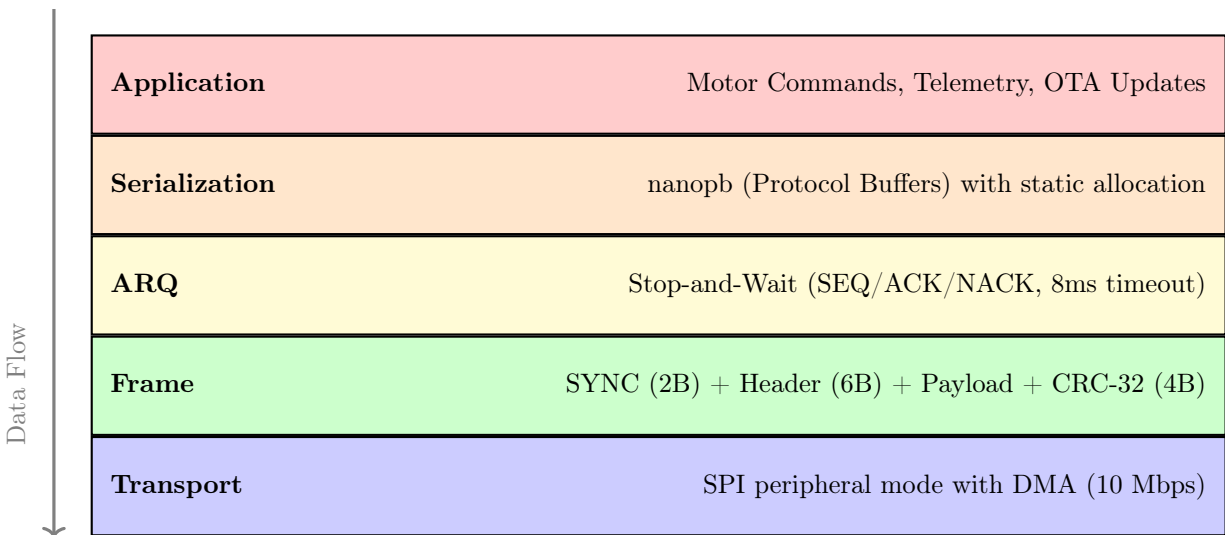


System Constraints:

- ESP32-S3-WROOM-1-N16: 16MB flash, 512KB SRAM, NO PSRAM
- 100Hz SPI communication (10ms period)
- 250Hz motor control loop (4ms period)
- Static allocation only (no malloc)

1.2 Key Design Decisions

1.2.1 Protocol Stack



1.2.2 Memory Budget

Table 1: SRAM Usage (512 KB total)

Component	Size (KB)	Usage (%)
DMA double-buffer	TBD	TBD
RX/TX buffers	TBD	TBD
Protobuf structs	TBD	TBD
ARQ state	TBD	TBD
Total	TBD	TBD

Table 2: Flash Usage (16 MB total)

Component	Size (KB)	Usage (%)
Generated protobuf	TBD	TBD
Nanopb runtime	TBD	TBD
Frame + ARQ layers	TBD	TBD
Total	TBD	TBD

TODO: Calculate and verify memory budget values for SRAM and Flash usage.

1.3 ARQ Communication Flow

1.3.1 Normal Operation (No Retransmission)

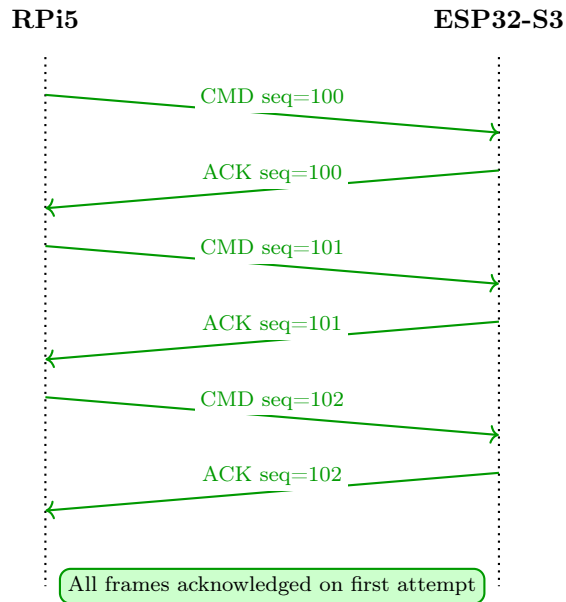


Figure 1: Normal ARQ Flow - Three Successful Transactions

1.3.2 Error Recovery Scenarios

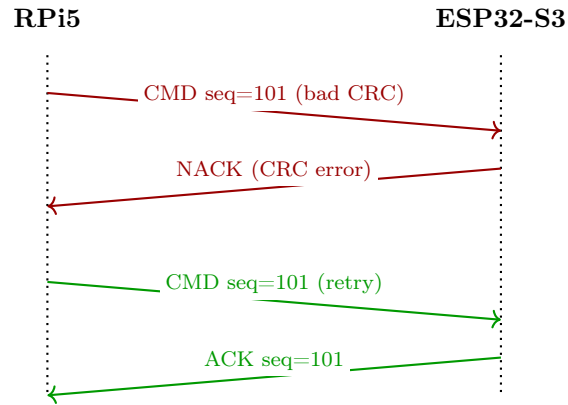


Figure 2: CRC Error - Retry with Same Sequence Number

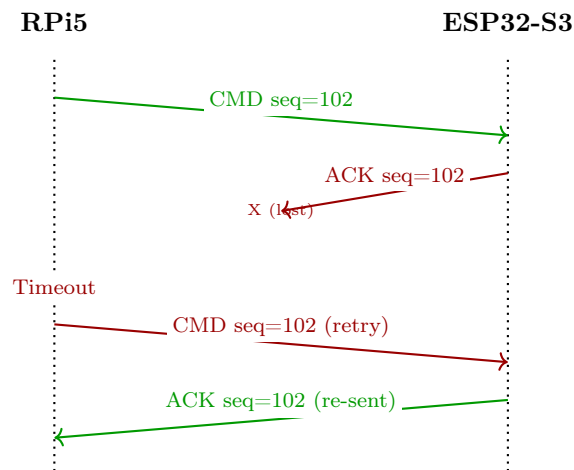


Figure 3: Duplicate Packet Scenario (Lost ACK)

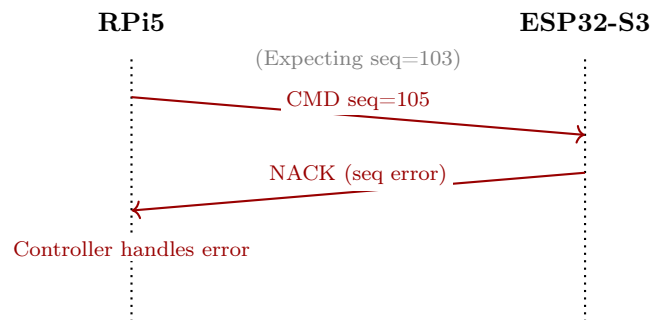


Figure 4: Out-of-Order Packet Scenario

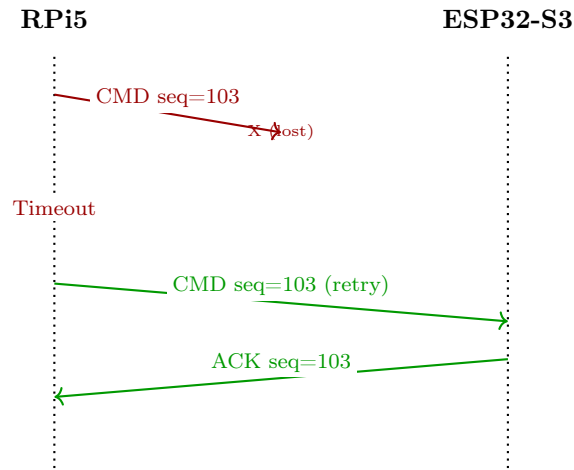


Figure 5: Lost Packet Scenario (Timeout)

1.4 Error Handling Strategies

CRC Mismatch

Action: NACK with CRC error reason
Recovery: Controller retry
Logging: Increment `crc_errors`

Sequence Error

Duplicate: Re-send ACK
Out-of-order: NACK + reject
Recovery: Controller retry

Timeout

RX timeout: Continue next cycle
500ms watchdog: Emergency stop
Retry fail: Log error

Invalid Protobuf

Action: NACK protocol error
Validate: IDs, ranges, offsets
Logging: Increment `proto_errors`